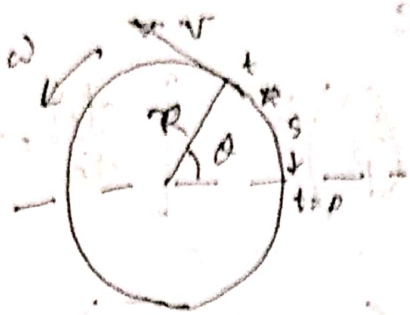
Summary of concepts & formula in Circular Motion

1.



In circular motion,

θ = angular displacement

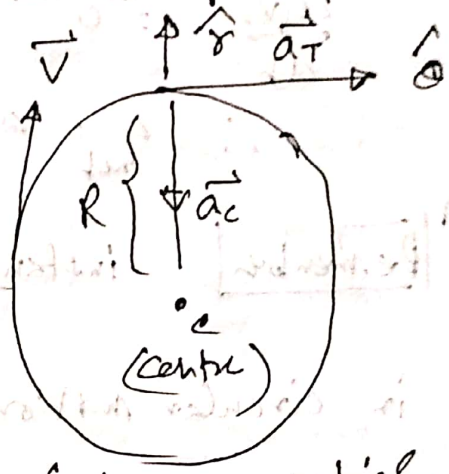
ω = angular ~~velocity~~ speed

v = linear ~~velocity~~, speed.

always tangential to circular path.

(i) $\boxed{\omega = \frac{d\theta}{dt}}$ (ii) $\boxed{v = \frac{ds}{dt}}$ (iii) rel. b/w v & ω $\boxed{v = \omega R}$

(2) In circular motion acceleration of the particle has two components



(i) tangental accl^n

$\boxed{a_T = \frac{d|\vec{v}|}{dt}}$

= rate of change of speed in circular motion

$\boxed{a_T = r\alpha}$, where

α = angular accl^n

$= \frac{d\omega}{dt}$ = rate of change of angular velocity.

$\hat{\theta}$ = tangential unit vector

$\hat{r}$ = radial unit vector.

(ii) radial or centripetal accl^n $a_c = \frac{v^2}{R} = \omega^2 R$.

(iii) $\because$ a_T & a_c are mutually perpendicular.

So net accl^n $a = \sqrt{(a_T)^2 + (a_R)^2}$

OR $= \sqrt{\left(\frac{d|\vec{v}|}{dt}\right)^2 + \left(\frac{v^2}{R}\right)^2}$

OR $= \sqrt{(r\alpha)^2 + \left(\frac{v^2}{R}\right)^2}$.

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(iv) in vector form,

$$\vec{a}_{net} = \vec{a}_T + \vec{a}_c$$

$$= \frac{d|\vec{v}|}{dt}\hat{\theta} - \left(\frac{v^2}{R}\right)\hat{r} \quad \left[\because \frac{d|\vec{v}|}{dt} = r\alpha\right]$$

<u>Note</u> $\vec{a}_c \cdot \vec{v} = 0 \quad (\because \vec{a}_c \perp \vec{v})$

(v) In uniform circular motion. ie circular motion with constant speed $|\vec{v}|$ = constant.

$\frac{d|\vec{v}|}{dt} = 0$ ie $a_T = 0$

$\therefore a_{net} = a_c = \frac{v^2}{R}$ or $\omega^2 R$

(3) $\boxed{\text{Remember}}$ instantaneous Power

$$P = \vec{F} \cdot \vec{v}$$

in circular motion $P = (\vec{F}_c + \vec{F}_T) \cdot \vec{v}$

[$|\vec{F}_c|$ = centripetal force = Sum of all forces (may be T, N, electric/magnetic, force, friction etc) toward centre. ⇒ this force changes dir. of velocity in circular motion. = mass (m) × centripetal acc^n $\left(\frac{v^2}{R}\right)$

As $\vec{F}_c \perp \vec{v}$ ∴ Power of centripetal force $P_c = \vec{F}_c \cdot \vec{v} = 0$,

i.e centripetal force is a no work force, it can not change KE and speed of particle]

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(3) So Power in circular motion

$$P = \vec{F_c}\cdot\vec{v} + \vec{F_T}\cdot\vec{v}$$

$$= 0 + \vec{F_T}\cdot\vec{v}$$

$$\boxed{P_{inst} = \vec{F_T}\cdot\vec{v}}$$

where $|\vec{F_T}|$ = tangential force = Force which changes magnitude of velocity i.e speed $|\vec{v}|$ in circular motion

$= m\,a_T = m\dfrac{d|\vec{v}|}{dt} = m\alpha r.$

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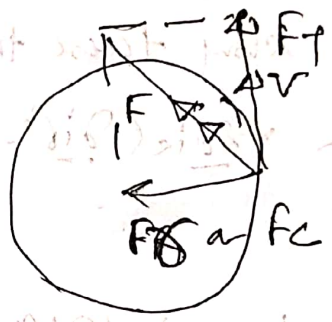
(4) So net force in circular motion

$$F = \sqrt{(F_T)^2 + (F_R)^2}$$

or

$$= \sqrt{(F_T)^2 + (F_c)^2}$$



[as $F_T \perp F_c$]

SO Know

In uniform circular motion where speed $|\vec{v}|$ $= v = \omega r$ = constant, $\dfrac{d|\vec{v}|}{dt}$ i.e $a_T = 0$, hence $F_T = m a_T = 0$, There is only centripetal force. F_c or $F_r = \dfrac{mv^2}{R}$

But in non-uniform circular motion where speed v $(=\omega r)$ changes, we have two forces. F_c (centripetal force = changes dir of velocity) & F_T (tangential force = changes speed)

So $\vec{F_{net}} = \vec{F_T} + \vec{F_R}$, $(\vec{F_T} \perp \vec{F_c})$.

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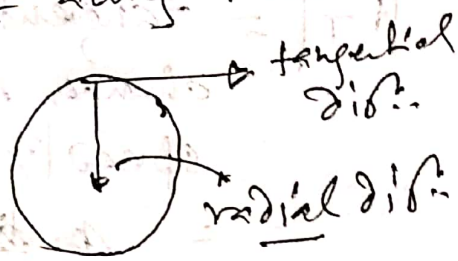
To solve circular motion problem:

Step 1. Draw free body diagram, ie draw the forces acting on the body

Step 2 Resolve the forces [if required] along tangential dir. & along radial dir. (to centre)



Step 3 Apply Newton laws separately along these two dir.

in radial dir., Net force to centre (ie centripetal) $= m \cdot \frac{v^2}{R}$ or $m\omega^2 R$

in tangential dir. Net force along tangent (ie tangential force) $= ma_T$ ie $m\frac{d|\vec{v}|}{dt}$ or $mR\alpha$

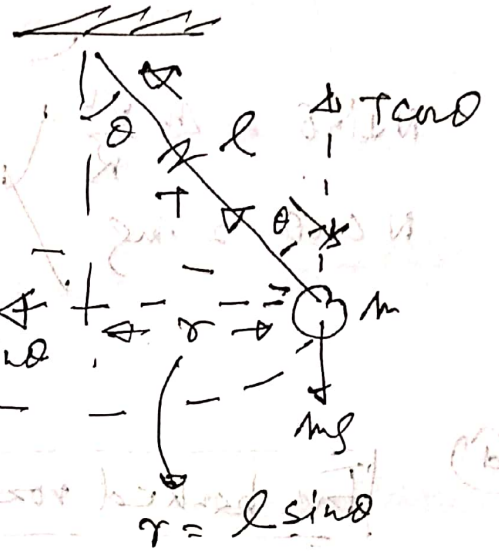
In addition to these in vertical circular motion, you have to apply conservation of mechanical energy, to find speed at one point if speed

Some imp problem types.

(1) Circular motion in horizontal plane
[speed $v = \omega r$ = constant]

(a) **CONICAL PENDULUM TYPE** :→

pendulum bob rotated in horizontal circle.



$T\cos\theta = mg$ — (1)

$T\sin\theta = m\frac{v^2}{r}$

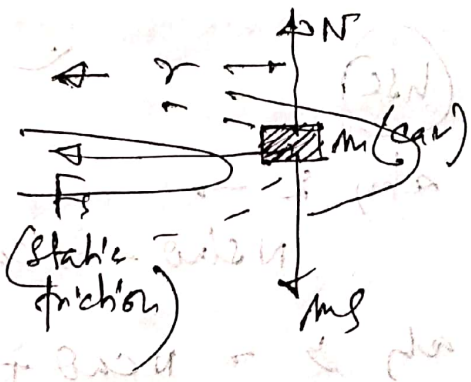
or

$m\omega^2 r$

remember $\boxed{r = l\sin\theta}$

(b) **MOTION OF A CAR IN HORIZONTAL CIRCULAR TURN** :

static friction (F_s) provides necessary centripetal force,



$F_s = \frac{mv^2}{r}$ — (1)

& $N = mg$ — (2)

∵ $F_s \le F_{smax}$ ie μN

∴ Condn for no skidding,

$\frac{mv^2}{r} \le \mu N$ ie μmg

$v^2 \le \mu g r$

$\boxed{v \le \sqrt{\mu g r}}$

∴ V_{max} for safe turn $= \sqrt{\mu g r}$.

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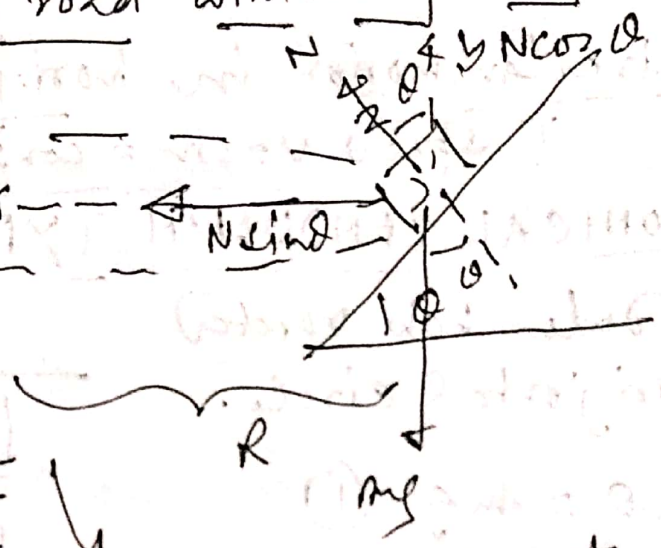
(c) **For banked road without friction :**



$x \quad a_c = \frac{v^2}{R}$

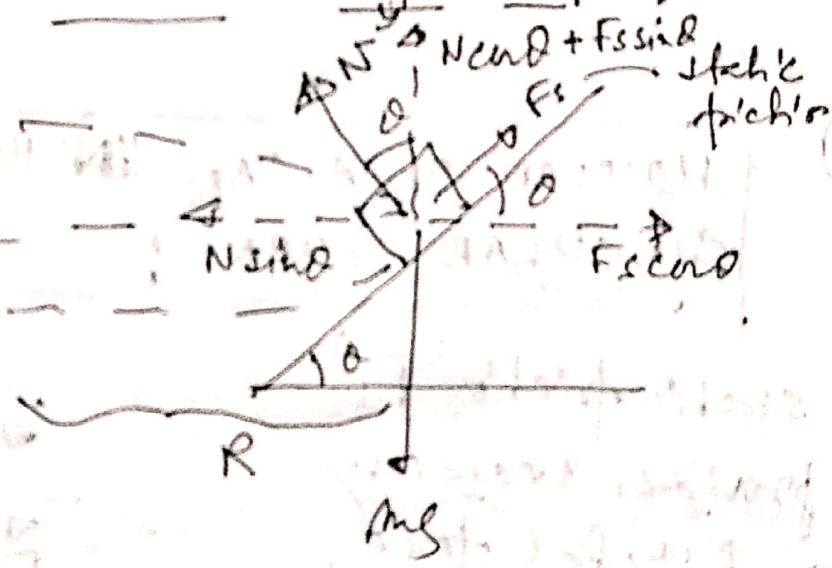
$N\sin\theta = m\frac{v^2}{R}$

$N\cos\theta = mg$

$\therefore \tan\theta = \frac{v^2}{rg}$

or $v = \sqrt{rg\tan\theta}$.

(d) **For banked road with friction :**



$a_c = \frac{v^2}{R}$

(hse)

along x →

$N\sin\theta - F_s\cos\theta = m\frac{v^2}{R}$ — (1)

along y → $N\cos\theta + F_s\sin\theta = mg$. — (2)

Put $F_s = F_{s\,max} = \mu N$ when $v = v_{max}$?

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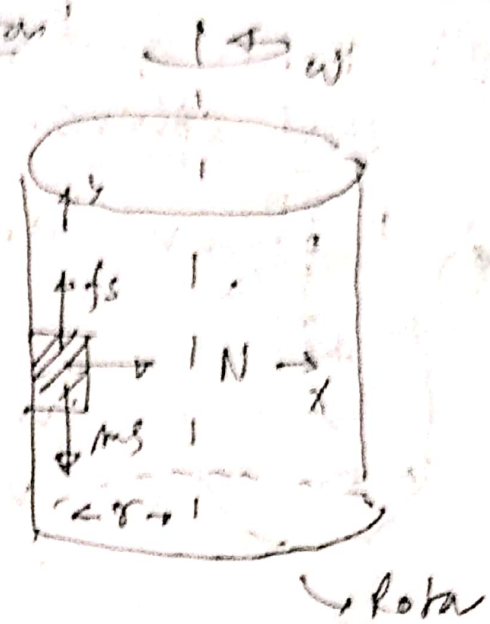
(e) 'Death Well' or 'Rotor'

If body (mass m) is at rest & in contact with the inner wall of 'Rotor' of radius R



then:

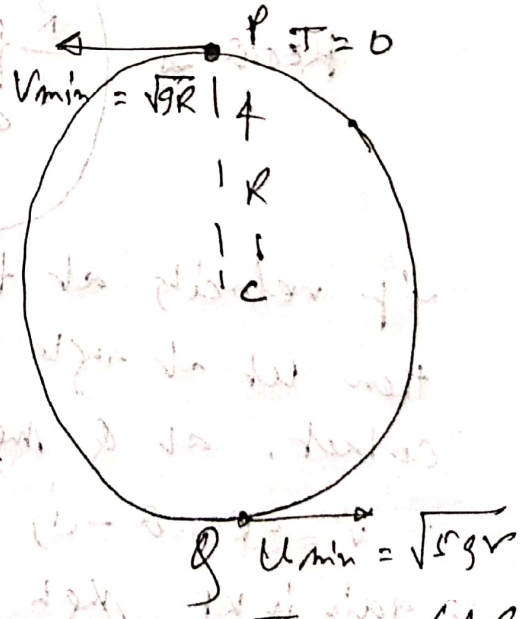
fs (static friction) balances weight

$$f_s = mg \quad (1)$$

& Normal force 'N' provides centripetal force

$$N = m\frac{v^2}{r} \text{ or } m\omega^2 r \quad (2)$$

(f) Remember: for circular motion of a body of mass 'm' connected to a string (massless) of length 'r' in a vertical plane:



a) Condition for looping the loop tension in string at topmost pt P

$$T \geq 0$$

b) Tmin at P = 0, Vmin at P = $\sqrt{gR}$

c) Vmin at Q = $\sqrt{5gr}$, Tmin at Q = 6mg.

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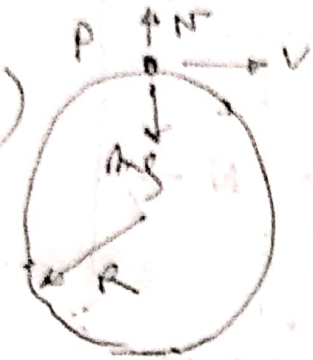
(g) Motion of a ball over a smooth solid sphere.

(i)



at topmost point (in fig P)

$mg - N = mv^2/R$

as $v \uparrow$ $N \downarrow$ & when $N = 0$,

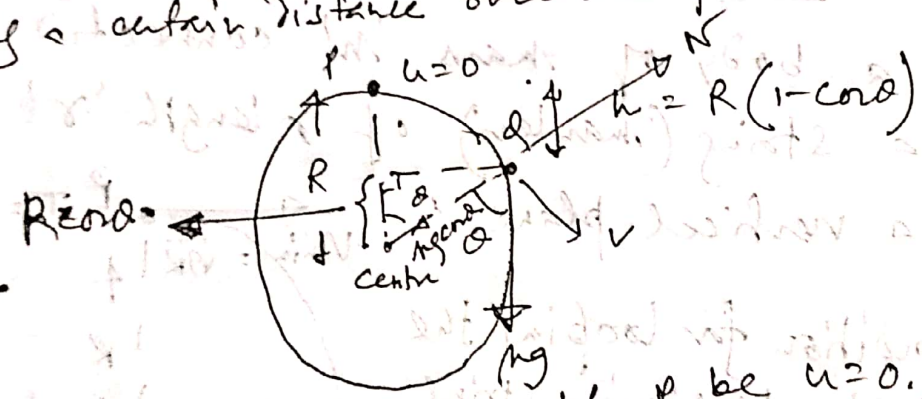
$v = v_{max}$.

$\therefore \frac{\cancel{m} v_{max}^2}{R} = \cancel{m} g$

$$\boxed{v_{max} = \sqrt{gR}}$$

From here we conclude that the ball will lose contact with the sphere right from the beginning if velocity of ball at topmost pt is $v > \sqrt{gR}$.

(ii) If $v < \sqrt{gR}$ it will lose contact after moving a certain distance over the sphere.



if velocity at topmost pt P be $u = 0$. then let at angle θ (in fig) the ball loses contact. at Q $mg\cos\theta - N = \frac{mv^2}{R}$ $\because N = 0$

$\therefore v^2 = gR\cos\theta$ — (i) from P → Q loss in G.P.E = gain in K.E, $mgh = \frac{1}{2}mv^2$

$\therefore \cancel{g}\cancel{R}(1-\cos\theta) = \frac{1}{2}\cancel{g}\cancel{R}\cos\theta \Rightarrow \frac{3}{2}\cos\theta = 1$

$\theta = \cos^{-1}\left(\frac{2}{3}\right)$.

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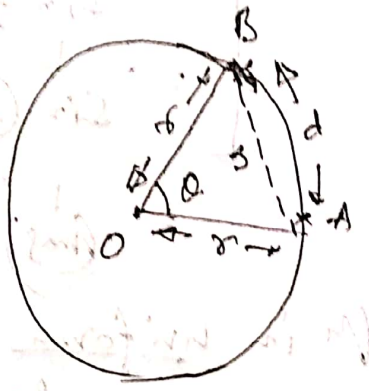
(1.p) problem type on kinematics of circular motion :-

> Ratio of average speed to magnitude of average velocity for any angular distance 'θ' in a circular motion in a given interval of time is $\frac{\theta}{2}\operatorname{cosec}\left(\frac{\theta}{2}\right)$

Explanation :-

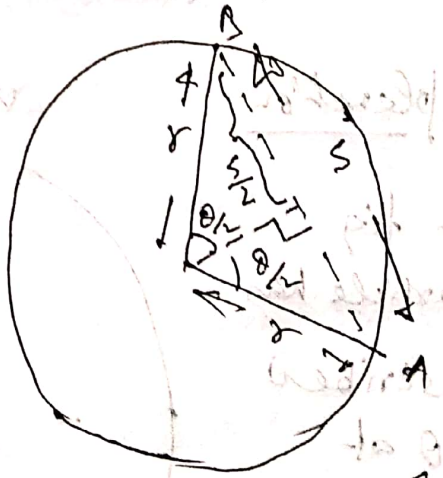


In fig particle has moved from pt A to pt B on circle of radius r in time t, describing an angle θ.

(i) Distance covered in time t is d = length of ARC AB $\therefore$ $\boxed{d = r\theta}$

$\therefore$ average speed $v = \frac{d}{t} \Rightarrow \boxed{v_{avg} = \frac{r\theta}{t}}$ — (1)

(ii) Displacement in time t is s = chord AB



from fig, $\frac{s/2}{r} = \sin\theta/2$

$\therefore \frac{s}{2} = r\sin\left(\theta/2\right)$

$\therefore \boxed{s = 2r\sin\left(\theta/2\right)}$

$\therefore$ Magnitude of average velocity $|\vec{V}_{avg}| = \frac{s}{t}$

$\therefore \boxed{|\vec{V}_{avg}| = \frac{2r\sin(\theta/2)}{t}}$ — (2)

(1) ÷ (2)

$\therefore \frac{d}{s}$ or $\frac{V_{avg}}{|\vec{V}_{avg}|} = \frac{\theta}{2} \times \frac{1}{\sin(\theta/2)} = \frac{\theta}{2}\operatorname{cosec}\left(\theta/2\right)$.

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Assignment prob

Example :- A particle travels two and a half revolutions of the circle of radius R in time 't'. The ratio of the average speed to the magnitude of average velocity in this time interval is

a) $\pi/2$ b) $\frac{5\pi}{\sqrt{2}}$ c) $\frac{5\pi}{2}$ d) $\frac{\pi}{5\sqrt{2}}$

$$\frac{V_{avg}}{|\vec{V}_{avg}|} = \frac{\theta}{2}\operatorname{cosec}\theta/2$$

$$= \frac{5\pi}{2} \times 1$$

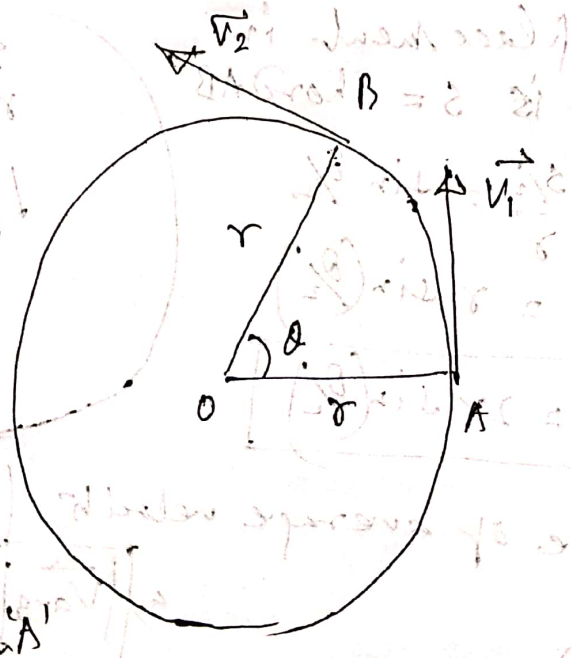
$$= \frac{5\pi}{2}$$

$$\left[\operatorname{cosec}\theta/2 = \frac{1}{\sin\theta/2} = \frac{1}{\sin\left(\frac{5\pi}{2}\right)} = 1 \text{ (ans)}\right]$$

To find average accln in uniform circular motion (v = constant)

Remember $\boxed{\vec{a}_{avg} = \frac{2v\sin\left(\frac{\theta}{2}\right)}{\Delta t}}$

Explanation



in fig particle has described ∡θ at centre O in time 'Δt' & come to 'B' from 'A'

$|\vec{V_2}| = |\vec{V_1}| = v$ (constant speed.

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change in velocity vector $\vec{\Delta v} = \vec{v_2} - \vec{v_1}$

$\vec{v_1} \perp OA$ & $\vec{v_2} \perp OB$, so as $\angle$ betn OA & $OB = \theta$ so angle betn $\vec{v_2}$ & $\vec{v_1}$ is also θ.

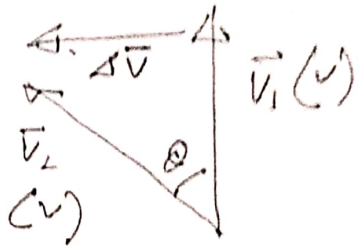
$$|\vec{\Delta v}| = \sqrt{v_2^2 + v_1^2 - 2v_1v_2\cos\theta}$$

$$= \sqrt{v^2 + v^2 - 2v^2\cos\theta}$$

$$= v\sqrt{2(1-\cos\theta)}$$

$$= v\sqrt{2 \times 2\sin^2\frac{\theta}{2}}$$

$$= 2v\sin\theta/2$$



$\therefore$ Average acc^n in uniform circular motion (const speed v) in time Δt is

$$|\vec{a_{avg}}| = \frac{|\vec{\Delta v}|}{\Delta t}$$

$$\boxed{|\vec{a_{avg}}| = \frac{2v\sin\theta/2}{\Delta t}}$$

problem:- Find magnitude of average acc^n of the tip of second hand of a clock in 10 seconds. (length of sec hand $r = 0.1$ m)

$$|\vec{a_{avg}}| = \frac{2v\sin\theta/2}{\Delta t},$$

$$= 2(\pi/300)\frac{\sin 30^\circ}{10}$$

$$= \frac{\pi}{3000} \text{ m/s}^2.$$

$v = \omega r$

$= \frac{2\pi}{T} r$

$= \frac{2\pi}{60} \times 0.1 = \frac{\pi}{300}$ m/s.

$\Delta t = 10$ sec

$\theta = \frac{2\pi}{60} \times 10 = 60^\circ$

[1 sec $= 6^\circ$ $\therefore$ 10 sec $= 60^\circ$]

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